

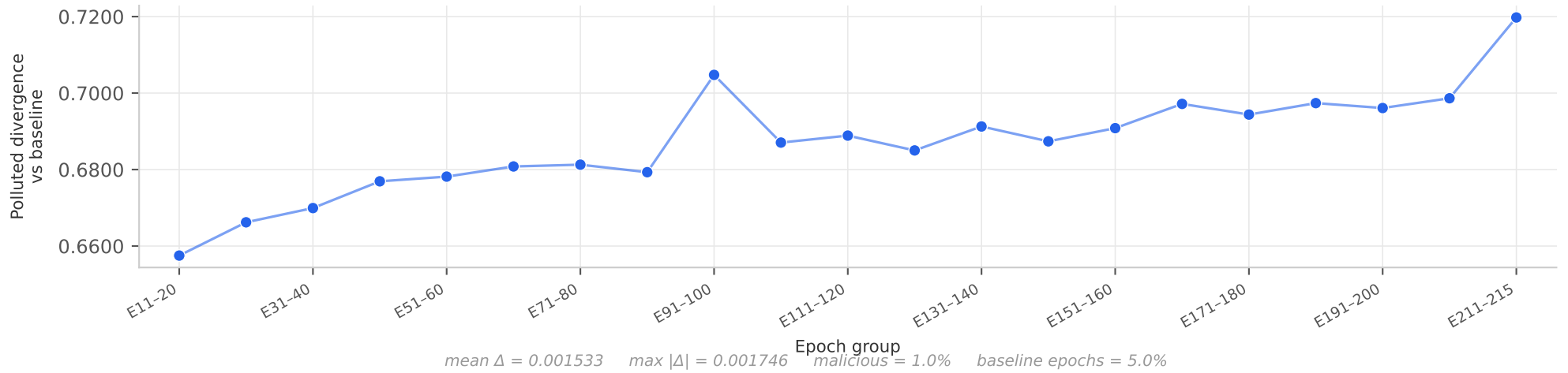
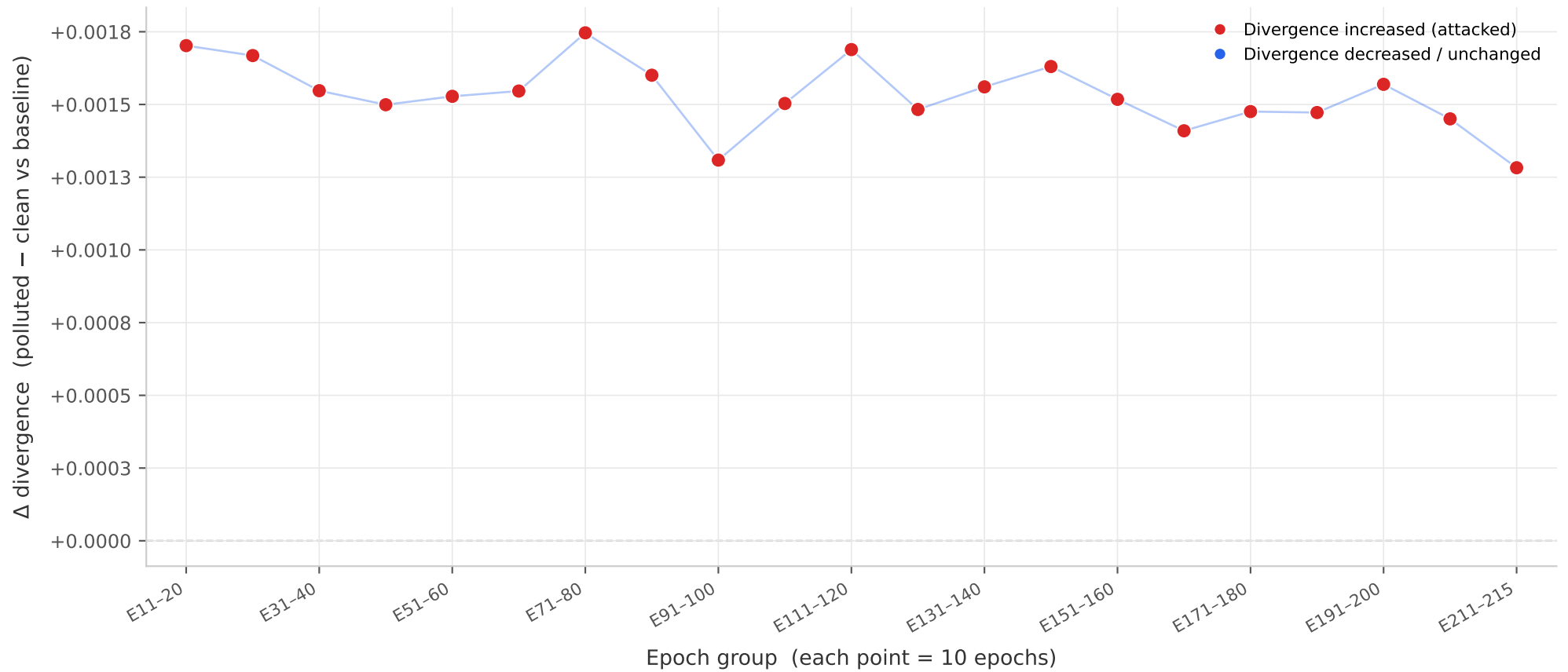
JS / KL Divergence Analysis

Epoch-wise Divergence from Baseline Distribution

Run Parameters		Baseline Distribution Summary	
Dataset	epochs_5.0_1.0.json	Total Flows	50558
Directory	CAIDA_Dataset/txtD280/epochs/	Unique Src IPs	14859
Timestamp	2026-02-19T18:59:35.801517	Unique Dst IPs	10585
Method	Jensen-Shannon + KL	Unique Src Ports	9030
Total epochs	215	Unique Dst Ports	20334
Baseline epochs	10 (5.0%)	Src IP Vocab Size	14859
Attack epochs	205	Dst IP Vocab Size	10585
Malicious traffic	1.0%	Src Port Vocab Size	9030
Epoch group size	10 epochs per point	Dst Port Vocab Size	20334
Metrics plotted	10		
Mode	FULL		

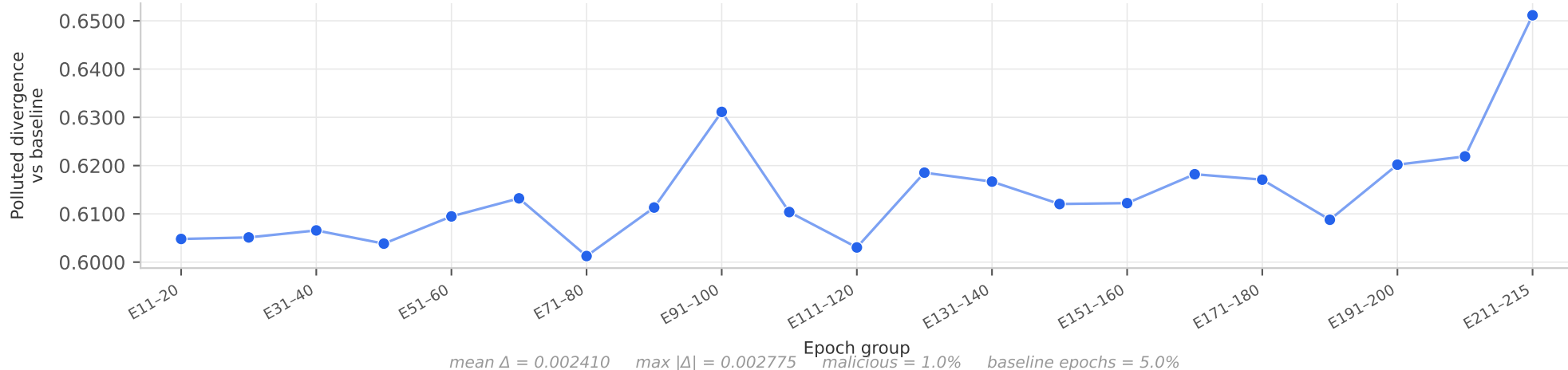
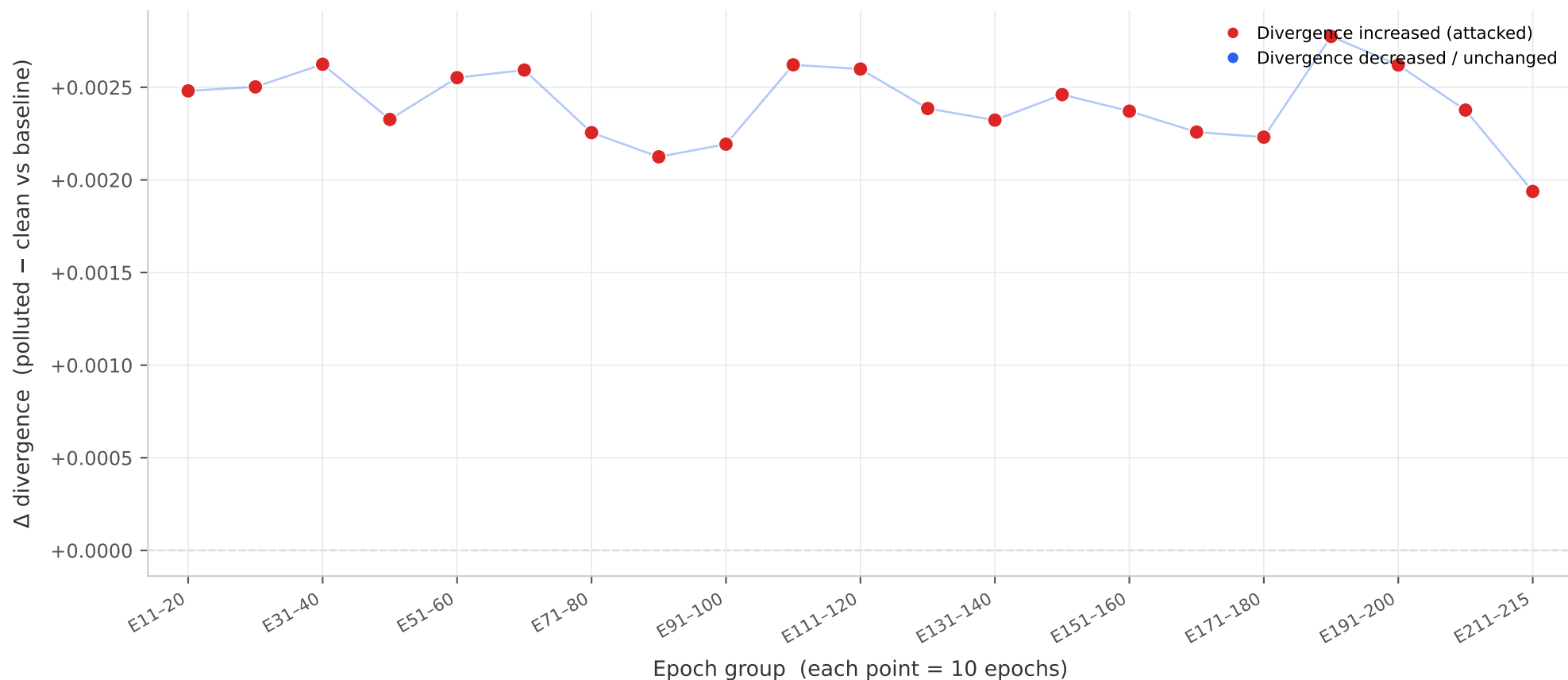
Source IP — JS Divergence

JS Divergence



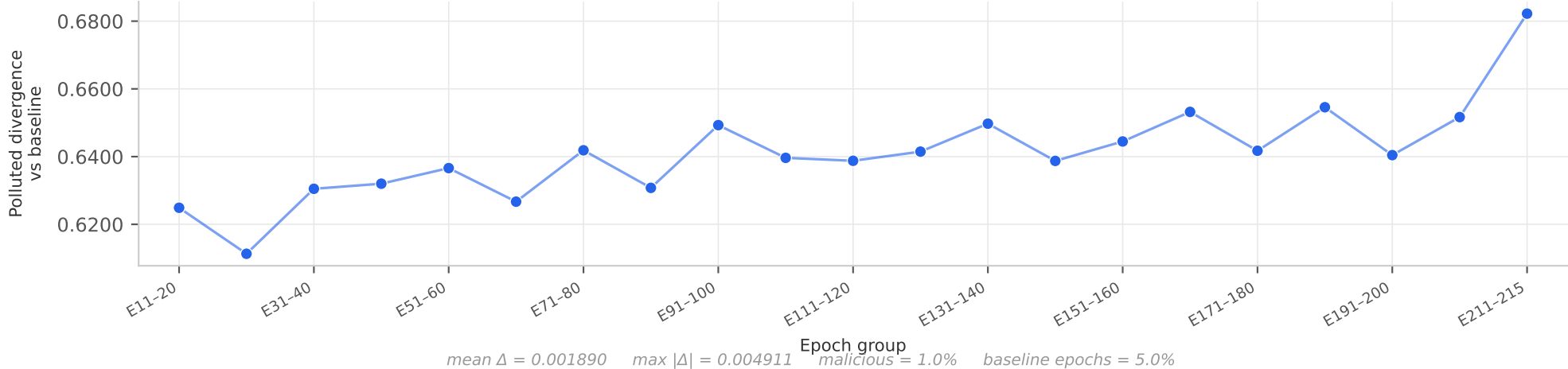
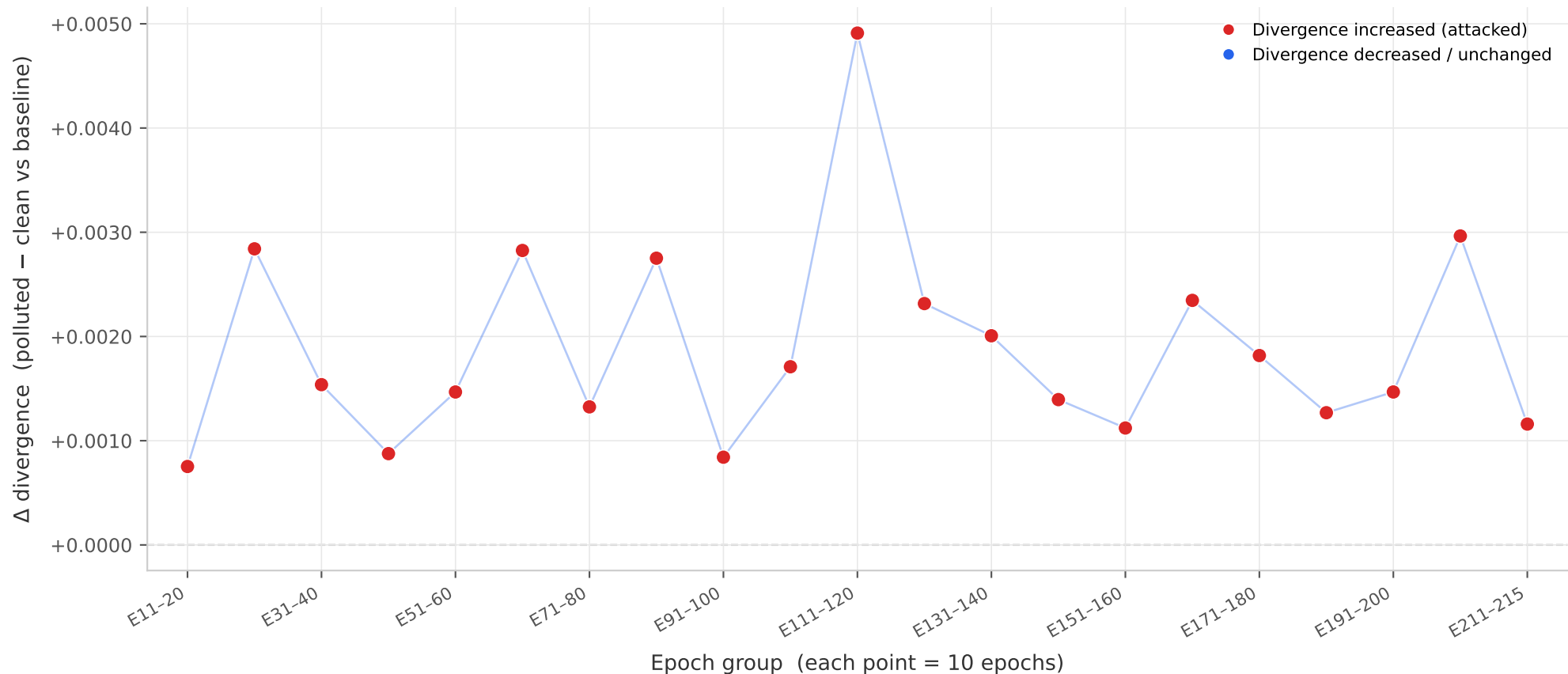
Destination IP — JS Divergence

JS Divergence



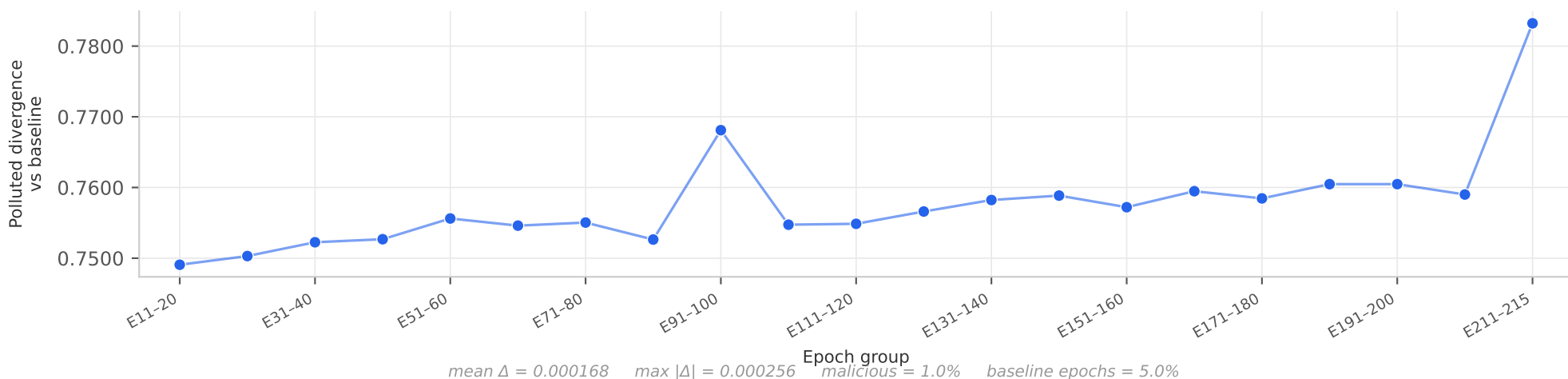
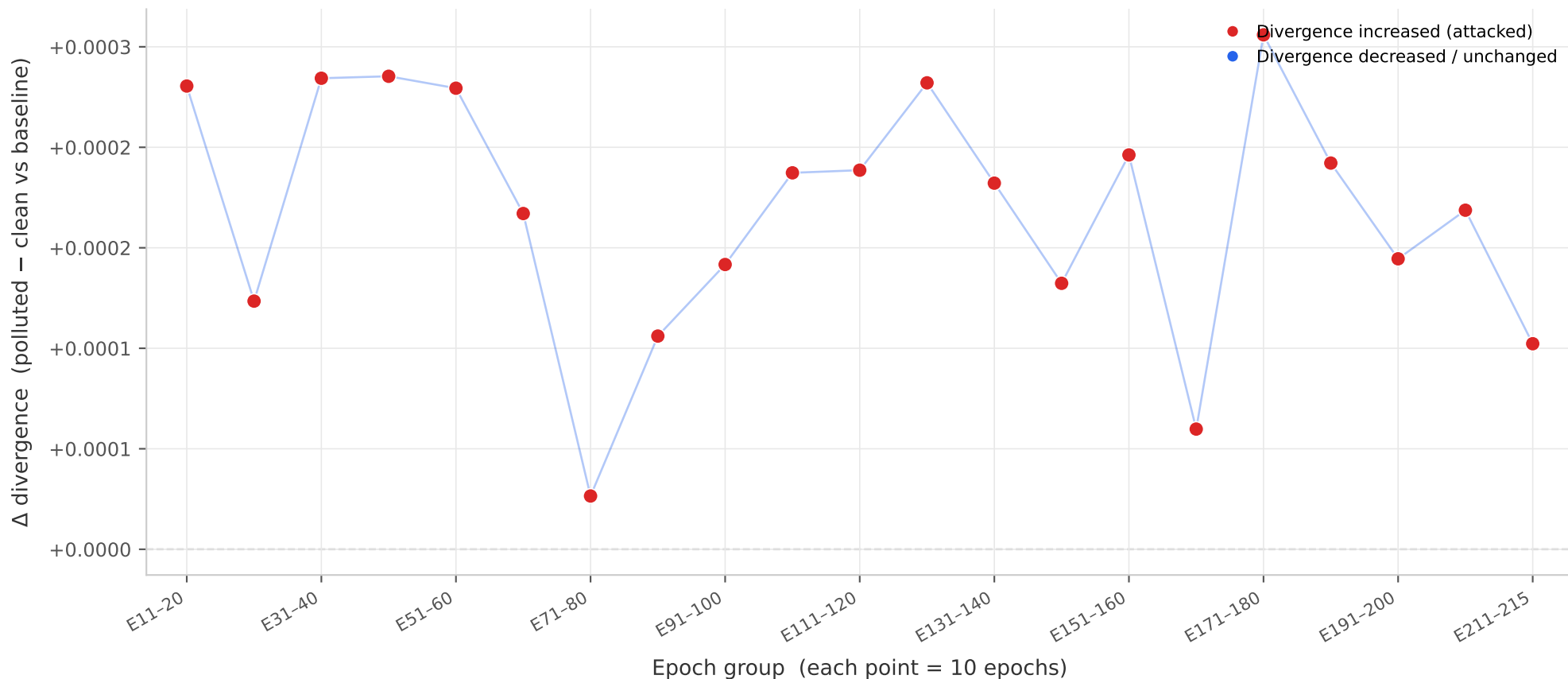
Source Port — JS Divergence

JS Divergence



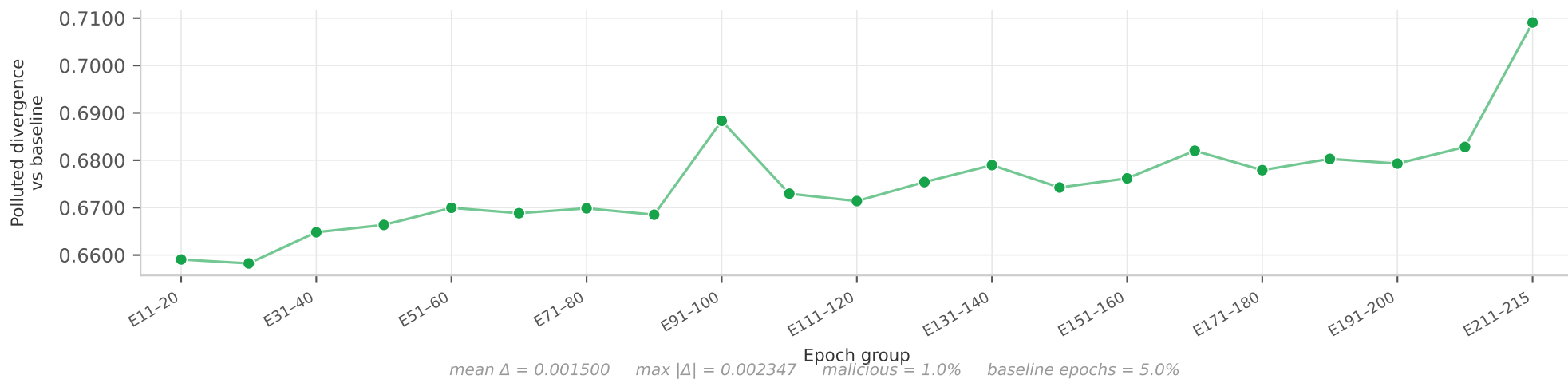
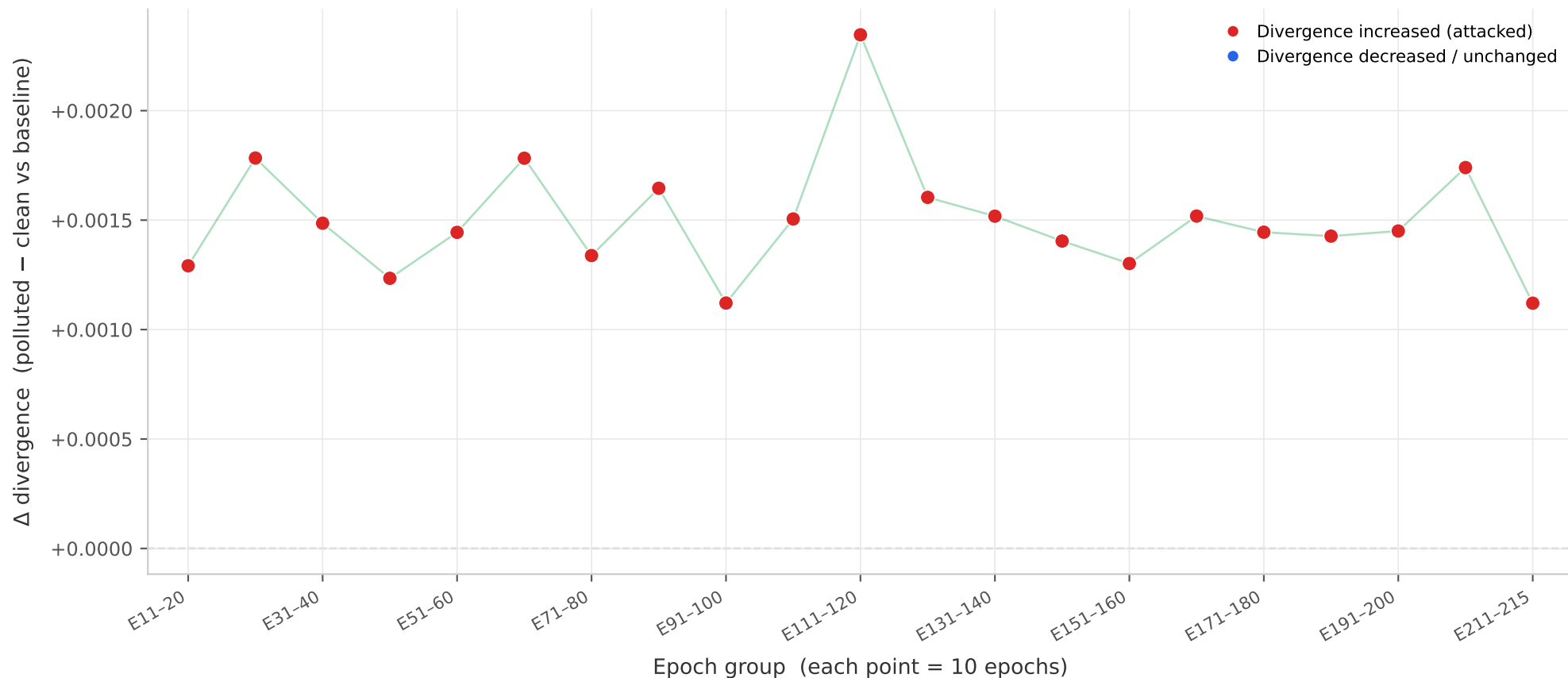
Destination Port — JS Divergence

JS Divergence



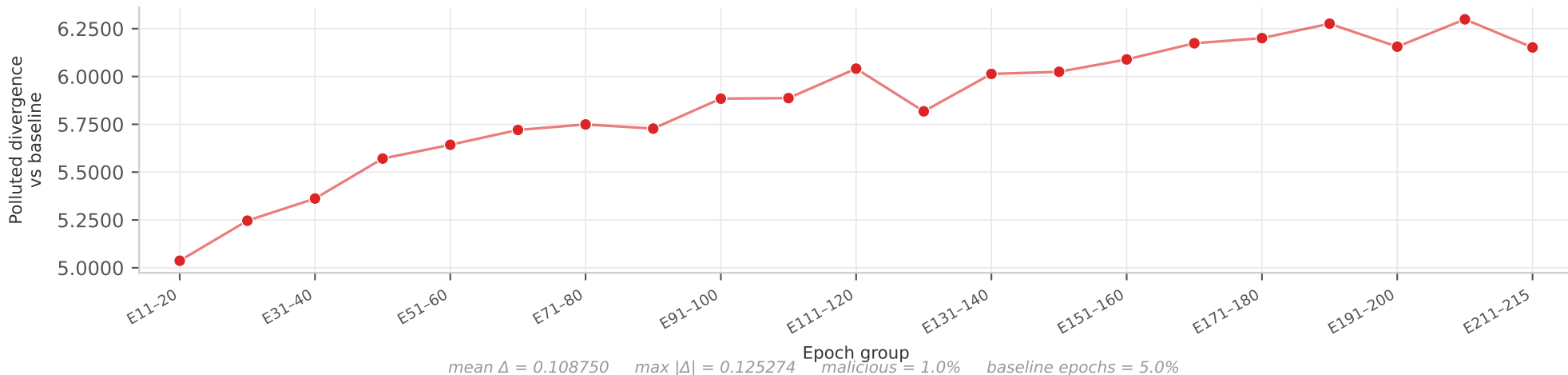
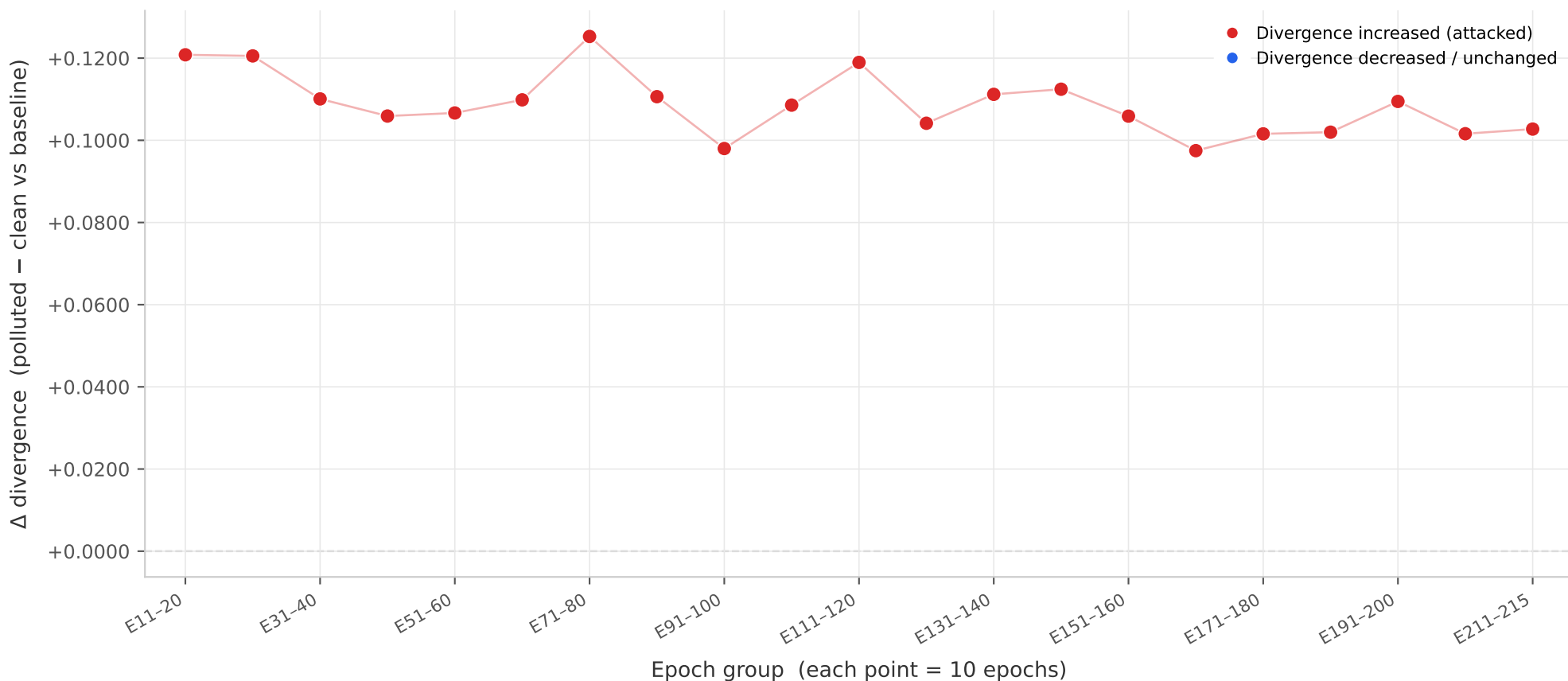
Combined (all features) — JS

Combined



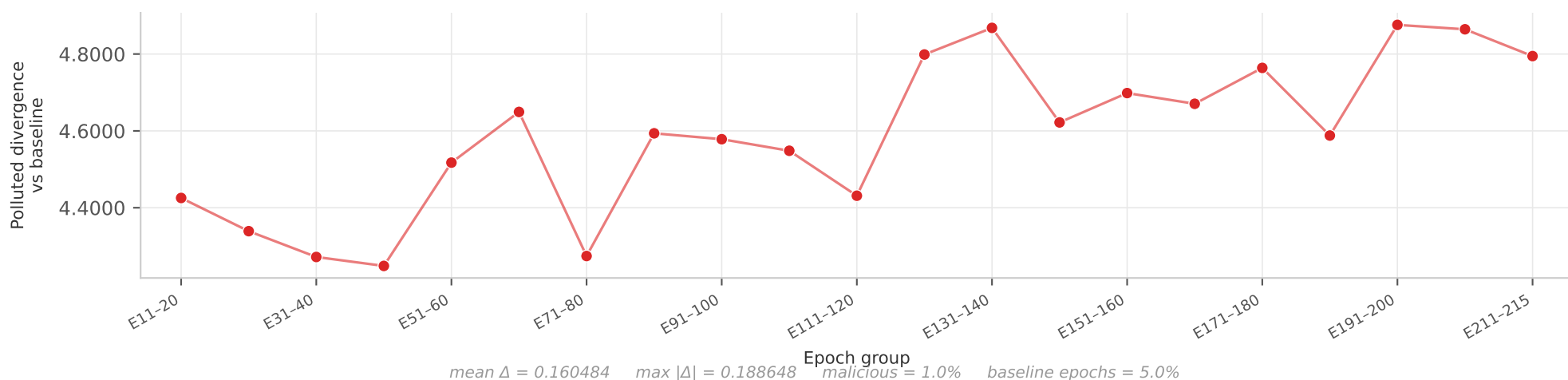
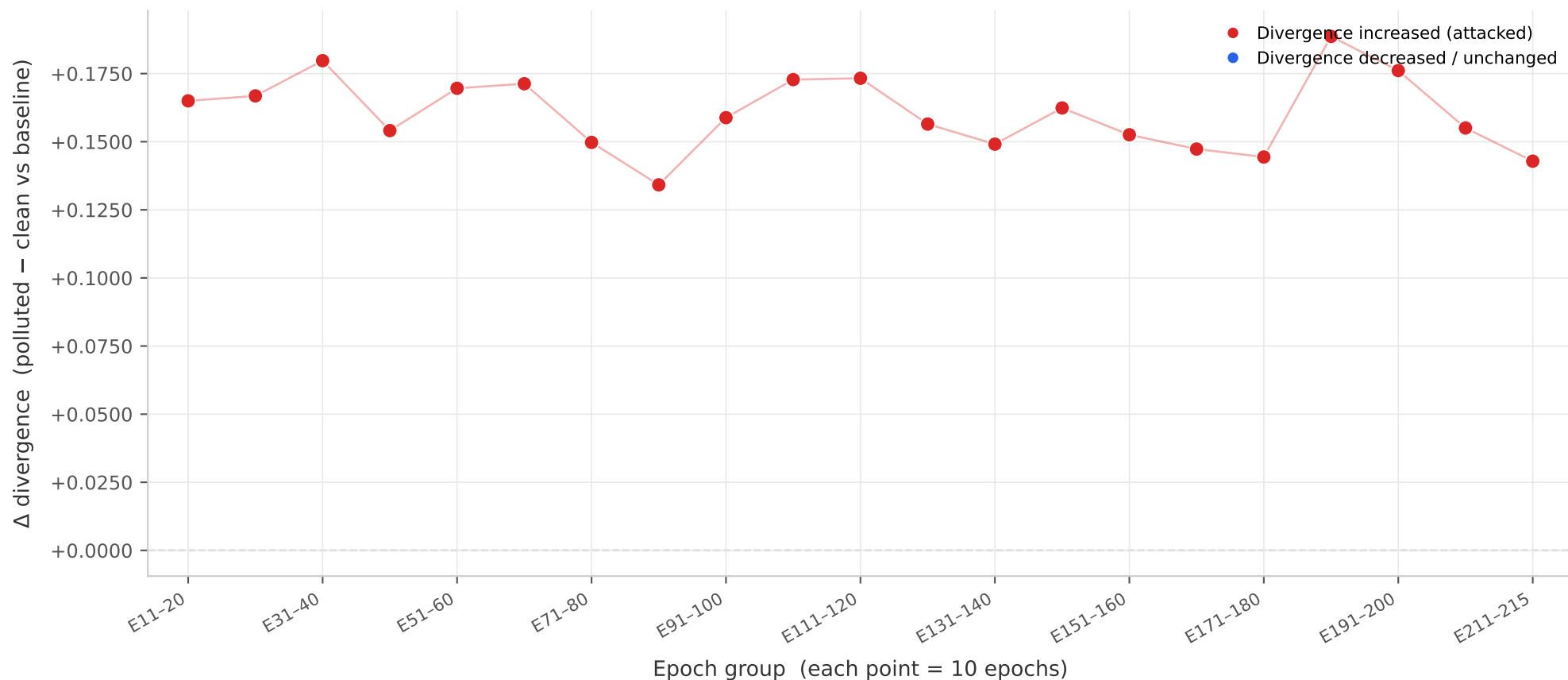
Source IP — KL Divergence

KL Divergence



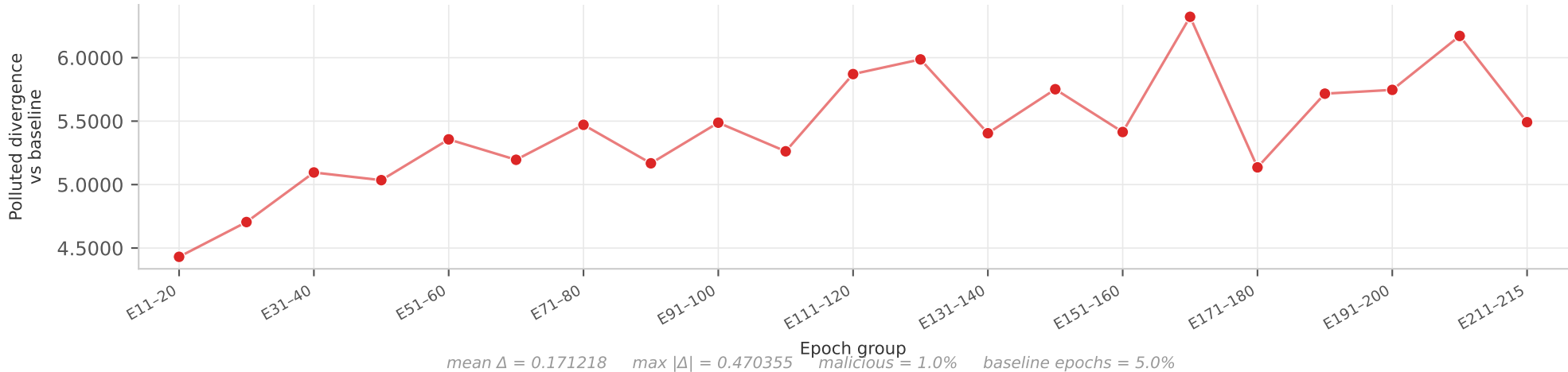
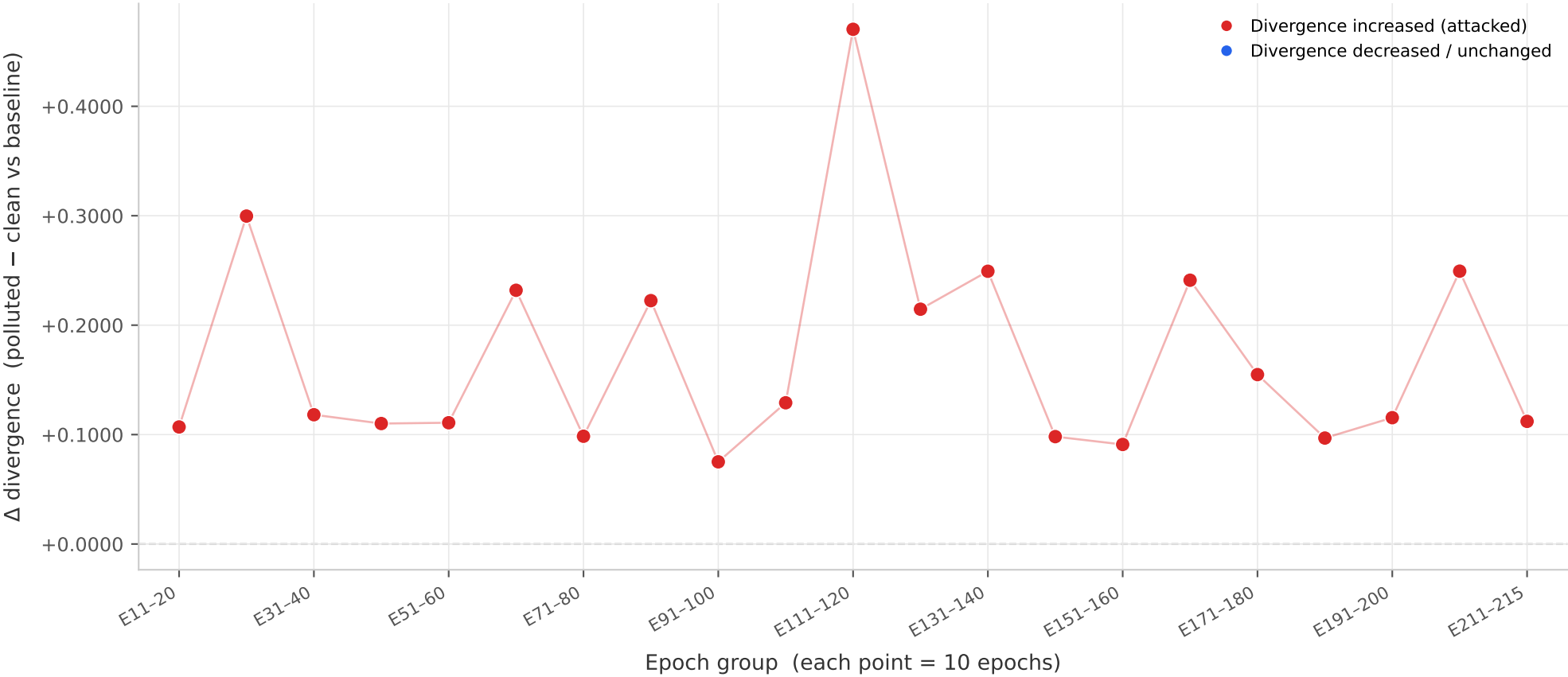
Destination IP — KL Divergence

KL Divergence



Source Port — KL Divergence

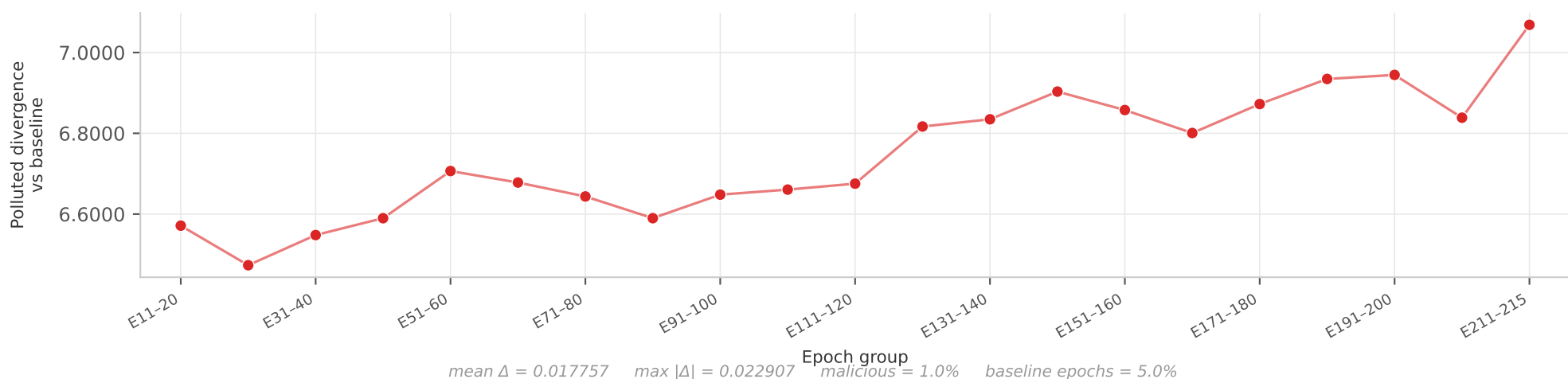
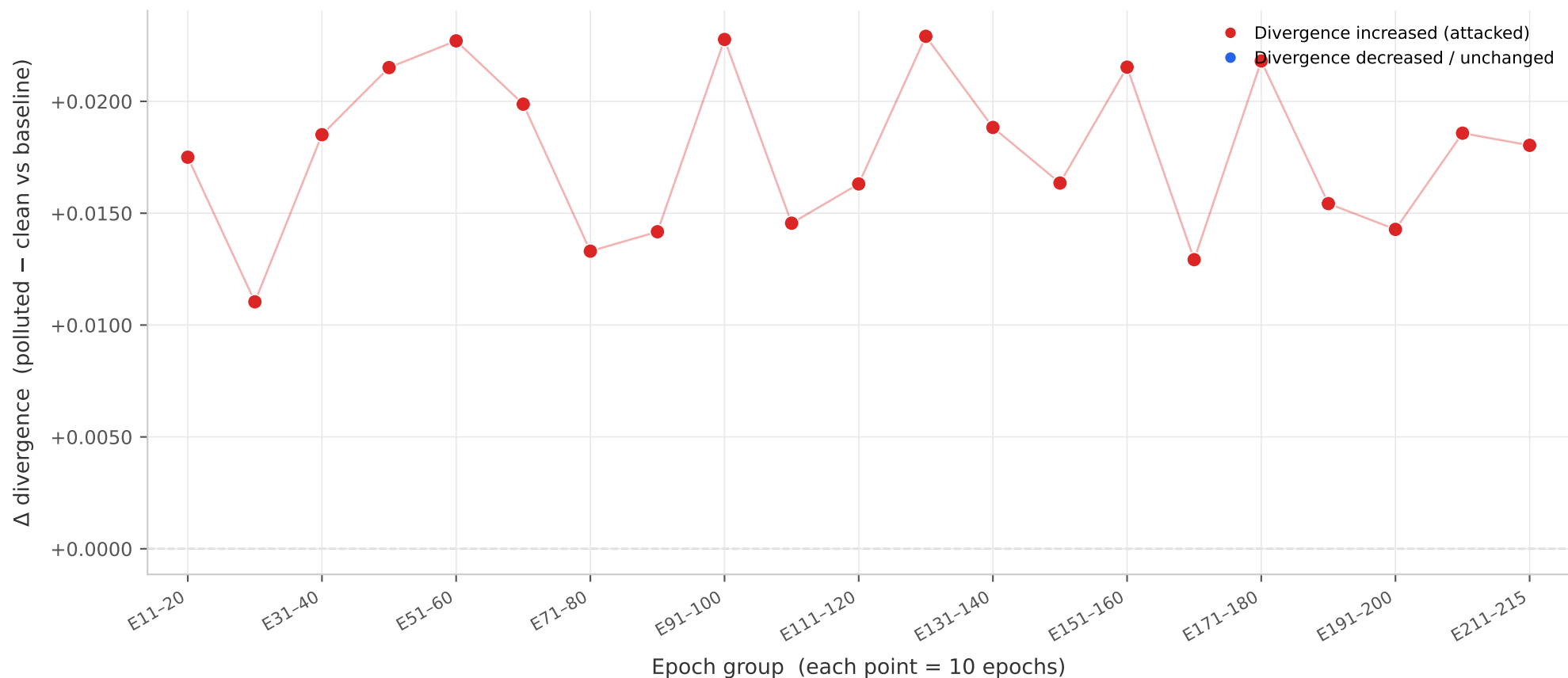
KL Divergence



mean Δ = 0.171218 max $|\Delta|$ = 0.470355 malicious = 1.0% baseline epochs = 5.0%

Destination Port — KL Divergence

KL Divergence



Total Flows

Flow

